

Nordic ID Sampo S2 One-Series

Nordic ID Sampo S2 One Series is a powerful fixed UHF RFID reader equipped with the high-end Nordic ID NUR2-1W module and an integrated antenna. With reading speeds up to 1000 tags/s and EU/US frequency support in one device, it suits POS, gates, and demanding integration use cases.



UHF RFID

Supported standard	ISO 18000-63 (EPC Class 1 Gen 2v2), AES authentication ISO/IEC 29167-10
Frequency band	ETSI 865.6–867.6 MHz and FCC/IC 902–928 MHz
Regulatory	CE ETSI EN 302 208, CE ETSI EN 301 489, FCC part 15.247 IC RSS-210
Max receive sensitivity	-81 dBm
Typical reading speed	Up to 1000 tags/s
Radiated power (integrated antenna)	Max 33 dBm (2 W) ERP / 3.3 W EIRP; Min 4 dBm ERP
Integrated antenna features	Wide band ETSI + FCC/IC, 7 dBic circular polarized, Beam width 80°, Nominal reading distance 10 m
Conducted power for external antenna ports	30 dBm (1 W)
External antenna port	3 pcs. 50 Ω / SMA Female
Isolation between external antenna ports	35 dB / typical

PLATFORM

CPU	1.2 GHz 64-bit Quad-Core Cortex-A53 processor
Operating system	Debian based embedded Linux
Memory	1 GB RAM / 4 GB flash

USER INTERFACE

USB	USB host, type A / USB device, type B, USB HID class supported
SIM	Mini-SIM
Indicators	4 pcs (programmable) LEDs; Buzzer; Capacitive sensor for triggering

CONNECTIVITY

GPIO	4 inputs / 4 outputs, opto isolated
Wireless WAN (optional)	3G HSPA / 2G GPRS / EDGE
LAN	Ethernet 10/100 Mbit
Wireless LAN (optional)	IEEE 802.11 a/b/g/n, dual band 2.4 / 5 GHz

POWER

External Power Supply	PoE 802.3af or AC/DC adapter: 100-240 VAC / 24 VDC, 1.25 A
Operating Power	12 W PoE, 20 W DC

SIZE AND WEIGHT

Dimensions	(W) 200 x (L) 260 x (H) 36 mm
Weight	810 g / 28.6 oz

ENVIRONMENT

Environmental sealing	IP20, for indoor use only
Operating Temperature	-20 to 55 °C
Storage Temperature	-40 to 85 °C
Mounting	VESA 75/100 Compatible

SOFTWARE INTERFACE

Data management	Nordic ID Radea
Device management	Nordic ID Radea, SOTI®
API support	NUR API for RFID and RESTful service
Other protocols	LLRP

FEATURE HIGHLIGHTS

- Powerful UHF RFID long-range reading performance
- Integrated Nordic ID NUR2-1W module for increased performance and reading speeds
- Integrated Linux based computer for versatile SW development
- Versatile host and device connectivity options
- Avoid cross readings due to high isolation between antenna ports
- One device for EU or US frequency environments

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed UHF RFID Reader